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The Effect of Sedation with Inhaled Anesthetics on Cognitive and Psychiatric Outcomes in Critically Ill Adults: A Systematic Review Protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-052893
Article Type:	Protocol
Date Submitted by the Author:	28-Apr-2021
Complete List of Authors:	Cuninghame, Sean; University of Western Ontario Department of Medicine, Medicine gorsky, kevin; University of Toronto, Anesthesiology and Pain Medicine Francoeur, Conall; Laval University, Department of Pediatrics Withington, Davinia; McGill University Faculty of Medicine, Anesthesiology Burry, Lisa; Sinai Health System, Departments of Pharmacy and Medicine; University of Toronto Jerath, Angela; Sunnybrook Health Sciences Centre, Anesthesia Slessarev, Marat ; Western University, Department of Medicine
Keywords:	Adult intensive & critical care < ANAESTHETICS, RESPIRATORY MEDICINE (see Thoracic Medicine), Delirium & cognitive disorders < PSYCHIATRY, COVID-19

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The Effect of Sedation with Inhaled Anesthetics on Cognitive and Psychiatric Outcomes in Critically Ill Adults: A Systematic Review Protocol

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ABSTRACT

Introduction: The COVID-19 pandemic has renewed interest in the use of inhaled anesthetics for sedation of ventilated critically ill patients. Preliminary data show that inhaled anesthetics reduce lung inflammation, time to extubation, and ICU length of stay compared to intravenous sedatives. However, the impact of inhaled anesthetics on cognitive and psychiatric outcomes is not well described in this setting. Randomized controlled trials are underway to establish if inhaled anesthetics affect these and other patient and health system outcomes. Our aim is to summarize the known effects of inhaled sedatives on cognitive and psychiatric outcomes.

Methods & Analysis: In this systematic review, we will use MEDLINE, EMBASE, and PsychINFO to identify studies from 1970 – 2021 that assessed cognitive and psychiatric outcomes in critically ill adult patients sedated with inhaled anesthetics. We will include case series, observational, cohort, and randomized controlled trials. We will exclude case studies due to the heterogeneity of reporting in these studies. For randomized controlled trials comparing inhaled to intravenous sedation, we will report cognitive and psychiatric outcomes for both study arms. Studies will be selected based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses checklist. Data will be extracted from using a standardized data extraction tool by two independent reviewers. Studies will be assessed for bias using the Cochrane Risk-of-Bias tool for Randomized controlled trials, or the Newcastle-Ottawa Scale for cohort and case-control studies. Findings will be reported according to outcome and descriptive statistics will be used to illustrate findings in a narrative fashion.

Ethics and dissemination: The systematic review uses published data and therefore does not require ethics approval. Results will be disseminated via publication in peer-reviewed journals and presentation at conferences related to the field.

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PROSPERO Registration Number: CRD4202123645

Study Strengths and Limitations:

- To the best of our knowledge, this is the first systematic review examining the effects on volatile anesthetics on cognition and psychiatric outcomes in critically ill adults
- Important review as cognitive and psychiatric outcomes are important patient-centred outcomes and a number of randomized controlled trials are currently underway examining the effect of inhaled anesthetics on physiologic, patient and health-system outcomes in patients with COVID-19 and non-COVID-19 respiratory failure.
- Limited exclusion criteria will allow for the maximum of papers on this topic to be included in our analysis
- We expect few, if any, randomized controlled trials specifically analyzing cognitive and psychiatric outcomes, and the review will likely be made up mostly of case series and observational studies

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INTRODUCTION

The SARS-CoV2 2019 (COVID-19) pandemic and surge of critically ill patients has renewed interest in the use of inhaled anaesthetics for sedation for those with severe respiratory failure requiring mechanical ventilation. Inhaled anaesthetics could address the shortages of intravenous sedative encountered during this pandemic because they can achieve the deep levels of sedation often required for patients with severe respiratory failure [1-4]. Inhaled anaesthetics have additional benefits including reduced lung inflammation, improved oxygenation, and are associated with shorter time to extubation and intensive care unit (ICU) length of stay [5-7]. These are important considerations as the pandemic has put stress on the availability of critical care resources in many parts of the world [8, 9]. Randomized controlled trials are currently underway to establish whether sedation with inhalational anesthetics of critically ill patients with COVID-19 and other causes of respiratory failure improves patient and health system outcomes in comparison to traditional intravenous sedatives (e.g., clinicaltrials.gov, NCT04415060). While these trials will help to determine whether inhaled anaesthetics improve mortality and shorten duration of ventilation in patients with hypoxic respiratory failure, they will also assess the impact of inhaled sedatives on the quality of life measures after recovery from critical illness.

Cognitive impairment is common among critically ill patients and represents a spectrum of dysfunction from acute delirium to long-term cognitive impairment following discharge from the intensive care unit [10]. ICU delirium is a well-established acute entity that is associated with increased mortality, prolonged mechanical ventilation and hospitalization, and increased health-care costs [11, 12]. Furthermore, the presence of delirium while in the ICU is a predictor of long-term cognitive disability, the incidence of which ranges from 10-58% across various studies [11, 13, 14]. While one systematic review did not find an association between sedative use and

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cognitive dysfunction in ICU survivors [15], more recent prospective cohort studies found that IV sedative exposure in the ICU is associated with worse long-term cognitive function [14, 16]. Similarly, evidence suggests that intravenous sedation may contribute to psychiatric morbidity in the form of depression, anxiety, and post-traumatic stress disorder (PTSD), which are also common amongst ICU survivors [17-20]. While an association has been found between intravenous sedatives (i.e., opioids, benzodiazepines, and propofol) and the development of cognitive and psychiatric morbidity, the impact of inhaled anesthetics on these outcomes has not been well described.

There is a clear association between critical illness and both acute and long-term cognitive dysfunction and psychiatric disturbances. Cognitive function and mental health are important determinants of academic and work success, levels of happiness, and life expectancy [21-24]. Given renewed interest in the use of inhaled anesthetics for sedation of critically ill patients, it is imperative to ascertain the effect of these inhaled sedatives on cognitive and psychiatric outcomes. By providing an up-to-date summary and analysis of existing literature, this systematic review will lay the foundation for future research in this emerging field.

AIMS

The aim of this systematic review is to evaluate the impact of inhaled anesthetics compared to intravenous sedatives on cognitive and psychiatric outcomes among mechanically ventilated critically ill adults. We will also investigate the methods used to measure and report cognitive and psychiatric morbidity in this setting.

METHODOLOGY

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The PICO strategy for this review is summarized in Table 1.

Types of Studies

Study designs will include case series, observational and cohort studies, and randomized controlled trials. Case reports will be excluded given the often heterogenous nature of reporting results. We will consult with previous systematic reviews that have investigated inhaled anesthetics use and assessed for cognitive performance after exposure while in the ICU to help identify primary literature on this topic. There is no restriction with respect to study sample size. Publication date will include studies published from 1970 until December, 2020. Only English-language studies will be used.

Types of Participants

Inclusion Criteria:

(1) Adult (>18 years old) patients

(2) Admitted to any intensive care unit including general medical-surgical, trauma and subspecialty ICUs (for example cardiovascular, burns and neuro-based units).

(3) Patients who received inhaled sedatives during their time in an intensive care unit. Treatment could be initiated in the ICU, or in the operating room as long as inhaled anesthesia was continued in the ICU thereafter.

(4) Treated with halothane, enflurane, isoflurane, sevoflurane or desflurane.

(5) Were assessed for cognitive and psychiatric (anxiety, depression, PTSD) outcomes using either subjective or objective tools at any stage during ICU admission or following ICU discharge.

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Exclusion Criteria:

(1) Pediatric patients

(2) Studies evaluating ether, trichloroethylene, methoxyflurane, xenon, nitrous oxide or other inhaled agents no longer relevant to modern clinical practice.

(3) Studies in language other than English

Types of Intervention

Interventions of interest will include exposure to inhaled anesthetics (i.e., sevoflurane, desflurane, isoflurane) for sedation in critically ill patients, and the concentration and duration of exposure to inhaled anesthesia.

Comparator

While we expect few randomized controlled trials on this topic, our comparator for this systematic review will be any intravenous sedatives if the study design includes one.

Types of Outcome

The primary outcomes of interest are subjective or objective measures of cognition, anxiety, depression, or PTSD. Secondary outcomes are methods and instruments used to measure cognitive, anxiety, depression, or PTSD outcomes (i.e., subjective assessment vs. objective testing; if objective, the specific test(s) used).

Search Strategy

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With the aid of our institute's clinical librarian, an initial search of Ovid MEDLINE will be done to identify relevant keywords in article titles and abstracts. These keywords will then be included in a final expanded search of the MEDLINE, EMBASE, PsycINFO, and the Cochrane Central Controlled Trials Register databases and a collection of their titles and abstracts uploaded into the online systematic review system, Covidence.

Study Selection

Two independent reviewers will screen each article's Title and Abstract for inclusion in the final data extraction based on the aforementioned study inclusion/exclusion criteria using Covidence. In order to be considered for full text review, screened articles will need to be identified for inclusion by both reviewers. Conflicts will be resolved by a third independent reviewer not involved in the initial screen. Full-text articles from the screening process will then be read for final inclusion in the review. A PRISMA flow diagram will be used to illustrate the number of studies screened, those chosen for inclusion, as well as reason(s) for exclusion from data extraction after full text review. This diagram will be generated using Covidence automatically after annotating specific studies for inclusion/exclusion.

Data Extraction

Data extraction will be performed independently by two reviewers using Covidence's data extraction tool. We have created a data extraction tool and will test the tool on two papers prior to finalizing tool's parameters. Conflicts or missing data during data extraction are automatically flagged and will be resolved by a third reviewer when necessary. Data will be extracted with the following data points of interest:

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i. **Study Characteristics** – Publication type, Country, Date of publication, study size, and setting (i.e., Medical, Surgical, Cardiovascular, Trauma Centre, etc.), industry sponsorship.

ii. **Population Characteristics** – Age, Sex, Admission Diagnosis, Measurement of Sedation, Overall mortality

iii. **Intervention Characteristics -** Inhaled or intravenous agent used; indication for sedation (postoperative cardiac, postoperative non-cardiac, non-surgical, or mixed), concentration (minimum alveolar concentration (MAC) or end-tidal concentration for inhaled if available), primary vs adjunct mode of sedation, total duration of exposure to inhaled anesthetic, mode of administration (i.e., Anesthesia Conserving Device (AnaConDa) vs other)

iv. **Comparator Characteristics –** Comparator(s) of interest include intravenous sedative used (e.g., Propofol, opioid, benzodiazepine, ketamine), dose of comparator(s) of interest used and age and sex of comparator group.

v. **Outcomes –** For each of cognitive function, anxiety, depression, and PTSD: tool(s) of measurement (subjective vs objective, and the specific tool (s) used (i.e., questionnaires, Mini Mental State Examination, Trails A, etc. for cognition; Hospital Anxiety and Depression or Patient Health Questionaire-9 for depression, etc. or a battery of tests for each outcome)), definition of impairment (scores on objective testing), timing of measurement (e.g., while in ICU, prior to hospital discharge; 3 months post-discharge, etc.), rates of outcome of interest (e.g., % with depressive symptoms)

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i. For cognitive function only: domain of cognition affected (e.g., executive function, language, etc.).

ii. For Depression, Anxiety, PTSD only: rates of use of psychiatric medications (e.g., antidepressants/anxiolytics, etc.) in follow-up.

Methodological Appraisal

Once we have selected studies for data extraction, each individual study will also be assessed for risk of bias based on publication type and validated screening tools. In order to assess selected studies for bias, we will use the Cochrane Risk of Bias tool for any randomized controlled trials, and report individual domains of bias as either "High" or "Low" risk of bias. To assess biases in case-control and cohort studies, we will use the Newcastle-Ottawa Quality Assessment Scale. For any case series that are included, we will use the Joanna Briggs Institute (JBI) Critical Appraisal tool for case series.

Data synthesis

We will perform a narrative description of included full-text studies for each of the four outcomes of cognition, depression, anxiety, and PTSD. When available, we will present each individual outcome of interest in table format with a descriptive statistical focus with categorical variables (e.g., sex, prevalence of quantitative impairment) expressed as a frequency of events, and quantitative variables from each study presented as medians. We do not anticipate a sufficient number of randomized controlled trials in this area to allow for a meta-analysis to be performed.

Sensitivity analyses

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If possible, a sensitivity analysis will be performed on the indication for sedation (cardiac post-operative patients vs non-cardiac postoperative patients vs non-operative), as well as for duration of sedation (for more than 24 hours or less than 24 hours) in order to ascertain the effect of these variables on the outcome of interest.

Patient and Public Involvement

As this is a systematic review protocol, patients and the public were not directly involved in the formation of the protocol. Patient outcomes will be derived from original articles when published in the systematic review.

FUNDING

Internal departmental support.

CONFLICTS OF INTEREST

The authors do not have any conflicts of interest (financial or non-financial) to report. AJ and MS have an ongoing clinical trial (clinicaltrials.gov, NCT04415060) investigating the use of inhaled anesthetics in the ICU.

AUTHOR CONTRIBUTIONS

MS and SC designed the project with assistance from AJ. SC drafted the initial manuscript. KG, CF, DW, LB, and AJ all assisted in project design and writing of the manuscript. All authors edited and approved the final submission.

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Table 1. Inclusion and Exclusion Criteria

POPULATION	Adult (>18 year-old) patients admitted to critical care units
INTERVENTION	Sedation with Inhaled Anesthetics
COMPARATOR	Intravenous sedation
OUTCOME	Short- and Long-term Cognition and Psychiatric outcomes

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24 April 2021

Dear Dr. Aldcroft,

Please find our systematic review protocol attached that is entitled "**The Effect of Sedation with Inhaled Anesthetics on Cognitive and Psychiatric Outcomes in Critically Ill Adults: A Systematic Review Protocol**" which we wish to submit for publication in *BMJ Open*. We feel this systematic review protocol is suitable for publication in *BMJ Open* as it describes an important upcoming systematic review relevant to the current COVID-19 pandemic. Based on the journal description, it would be an excellent fit for publication in *BMJ Open.*

As we describe in the body of the text, the COVID-19 pandemic has put significant strain on critical care resources around the world. Inhaled anesthetics offer a promising alterative to intravenous sedatives and preliminary studies suggest they may offer improved patient and health system outcomes with both COVID-19 and non-COVID-19 respiratory failure. In this systematic review, we will review existing literature and report the effect of inhaled anesthetics on cognitive and psychiatric outcomes in critically ill patients. While cognitive function and psychiatric health are important patient-centred outcomes, they are rarely considered in survivors of critical illness. Randomized trials are currently underway to help us understand how inhaled anesthetics affect cognition, and other patient and health systems outcomes. This review is important to help inform these and future studies on the existing evidence surrounding inhaled anesthetics and Cognitive and Psychiatric outcomes.

The protocol is original work that has not been previously published, nor is it under consideration with another journal as a whole or in part. The authors have no conflicts of interest to disclose.

Thank you for your consideration.

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Sincerely,

Marat Slessarev, MD, PhD

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BMJ Open

The Effect of Sedation with Inhaled Anesthetics on Cognitive and Psychiatric Outcomes in Critically Ill Adults: A Systematic Review Protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-052893.R1
Article Type:	Protocol
Date Submitted by the Author:	13-Oct-2021
Complete List of Authors:	Cuninghame, Sean; University of Western Ontario Department of Medicine, Medicine gorsky, kevin; University of Toronto, Anesthesiology and Pain Medicine Francoeur, Conall; Laval University, Department of Pediatrics Withington, Davinia; McGill University Faculty of Medicine, Anesthesiology Burry, Lisa; Sinai Health System, Departments of Pharmacy and Medicine; University of Toronto Jerath, Angela; Sunnybrook Health Sciences Centre, Anesthesia Slessarev, Marat ; Western University, Department of Medicine
Primary Subject Heading:	Intensive care
Secondary Subject Heading:	Anaesthesia, Mental health, Neurology, Research methods
Keywords:	Adult intensive & critical care < ANAESTHETICS, RESPIRATORY MEDICINE (see Thoracic Medicine), Delirium & cognitive disorders < PSYCHIATRY, COVID-19

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ABSTRACT

Introduction: The COVID-19 pandemic has renewed interest in the use of inhaled anesthetics for sedation of ventilated critically ill patients. Preliminary data show that inhaled anesthetics reduce lung inflammation, time to extubation, and ICU length of stay compared to intravenous sedatives. However, the impact of inhaled anesthetics on cognitive and psychiatric outcomes is not well described in this setting. Randomized controlled trials are underway to establish if inhaled anesthetics affect these and other patient and health system outcomes. Our aim is to summarize the known effects of inhaled sedatives on cognitive and psychiatric outcomes.

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PROSPERO Registration Number: CRD42021236455

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INTRODUCTION

The SARS-CoV2 2019 (COVID-19) pandemic and surge of critically ill patients has renewed interest in the use of inhaled anaesthetics for sedation for those with severe respiratory failure requiring mechanical ventilation. Inhaled anaesthetics could address the shortages of intravenous sedative encountered during this pandemic because they can achieve the deep levels of sedation often required for patients with severe respiratory failure [1-4]. Inhaled anaesthetics have additional benefits including reduced lung inflammation, improved oxygenation, and are associated with shorter time to extubation and intensive care unit (ICU) length of stay [5-7]. These are important considerations as the pandemic has put stress on the availability of critical care resources in many parts of the world [8, 9]. However, there may be practical limitations to using inhaled anesthetics in the ICU including equipment and properly trained personnel. In addition, some studies report various adverse effects of long-term inhaled anesthetics such as dose-dependent respiratory depression, hypotension, malignant hyperthermia, diabetes insipidus, and hepatitis, among others [9]. Therefore, current randomized controlled trials will be helpful in establishing whether sedation with inhalational anesthetics of critically ill patients with COVID-19 and other causes of respiratory failure is safe and improves patient and health system outcomes in comparison to traditional intravenous sedatives (e.g., clinicaltrials.gov, NCT04415060). While these trials will help to determine whether inhaled anaesthetics improve mortality and shorten duration of ventilation in patients with hypoxic respiratory failure, they will also assess the impact of inhaled sedatives on the quality-of-life measures after recovery from critical illness.

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increased mortality, prolonged mechanical ventilation and hospitalization, and increased health-care costs [11, 12]. Furthermore, the presence of delirium while in the ICU may be a predictor of long-term cognitive disability, the incidence of which ranges from 10-58% across various studies [11, 13, 14]. While one systematic review did not find an association between sedative use and long-term cognitive dysfunction in ICU survivors [15], some prospective cohort studies found that IV sedative exposure in the ICU is associated with worse long-term cognitive function [14, 16]. However, a recent study found that when comparing IV sedation with a non-sedation approach in critically ill adults, there was no difference in cognition at 3 months following ICU discharge [17]. There is also some evidence to suggest that intravenous sedation may contribute to psychiatric morbidity in the form of depression, anxiety, and post-traumatic stress disorder (PTSD), which are also common amongst ICU survivors [18-21]. While an association has been found between intravenous sedatives (i.e., opioids, benzodiazepines, and propofol) and the development of cognitive and psychiatric morbidity, the impact of inhaled anesthetics on these outcomes has not been well described.

There is a clear association between critical illness and both acute and long-term cognitive dysfunction and psychiatric disturbances. Cognitive function and mental health are important determinants of academic and work success, levels of happiness, and life expectancy [22-26]. Given renewed interest in the use of inhaled anesthetics for sedation of critically ill patients, it is imperative to ascertain the effect of these inhaled sedatives on cognitive and psychiatric outcomes. By providing an up-to-date summary and analysis of existing literature, this systematic review will lay the foundation for future research in this emerging field.

AIMS

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The aim of this systematic review is to evaluate the impact of inhaled anesthetics compared to intravenous sedatives on cognitive and psychiatric outcomes among mechanically ventilated critically ill adults. We will also investigate the methods used to measure and report cognitive and psychiatric morbidity in this setting.

METHODOLOGY

The PICO strategy for this review is summarized in Table 1.

Types of Studies

Study designs will include case series, observational and cohort studies, and randomized controlled trials. Case reports will be excluded given the often heterogenous nature of reporting results. We will consult with previous systematic reviews that have investigated inhaled anesthetics use and assessed for cognitive performance after exposure while in the ICU to help identify primary literature on this topic. There is no restriction with respect to study sample size. Publication date will include studies published from 1970 until December, 2020. Only English-language studies will be used.

Types of Participants

Inclusion Criteria:

(1) Adult (>18 years old) patients

(2) Admitted to any intensive care unit including general medical-surgical, trauma and subspecialty ICUs (for example cardiovascular, burns and neuro-based units).

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(3) Patients who received inhaled sedatives for any period of time during their admission to an intensive care unit. Treatment could be initiated in the ICU, the operating room, or the emergency department.

(4) Treated with halothane, enflurane, isoflurane, sevoflurane or desflurane.

(5) Were assessed for cognitive and psychiatric (anxiety, depression, PTSD) outcomes using either subjective or objective tools at any stage during ICU admission or following ICU discharge.

Exclusion Criteria:

(1) Pediatric patients

(2) Studies evaluating ether, trichloroethylene, methoxyflurane, xenon, nitrous oxide or other inhaled agents no longer relevant to modern clinical practice.

(3) Studies in language other than English

Types of Intervention

Interventions of interest will include exposure to inhaled anesthetics (i.e., sevoflurane, desflurane, isoflurane) for sedation in critically ill patients, and the concentration and duration of exposure to inhaled anesthesia. Patients who received adjunctive intravenous sedation (e.g., concomitant benzodiazepiens or opioids) will be included as long as they satisfy the above inclusion criteria.

Comparator

While we expect few randomized controlled trials on this topic, our comparator for this systematic review will be any intravenous sedatives if the study design includes one.

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Types of Outcome

The primary outcomes of interest are subjective or objective measures of cognition, anxiety, depression, or PTSD. Secondary outcomes are methods and instruments used to measure cognitive, anxiety, depression, or PTSD outcomes (i.e., subjective assessment vs. objective testing; if objective, the specific test(s) used).

Search Strategy

The search strategy will be performed and communicated in a way to maximize reproducibility of the search results and study conclusions following the PRISMA and PRISMA-S guidelines. An initial search of Ovid MEDLINE will be performed in order to identify relevant MeSH keywords in article titles and abstracts based on the PICO question outlined in Table 1. These keywords will then be included in a final expanded search of the MEDLINE, EMBASE, PsycINFO, and the Cochrane Central Controlled Trials Register databases and a collection of their titles and abstracts uploaded into the online systematic review system, Covidence. We will include articles from 1970 to 2021. In addition, the reference sections of each of the included studies will also be reviewed in order to ensure all relevant articles are included. This includes all items discussed in the newly published PRISMA-S checklist for reporting specifically on the literature search component of Systematic Reviews [27]. The full planned Search Strategy can be found in the Supplemental File of this manuscript as part of completing the PRISMA-S checklist.

Study Selection

Two independent reviewers will screen each article's Title and Abstract for inclusion in the final data extraction based on the aforementioned study inclusion/exclusion criteria using Covidence.

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In order to be considered for full text review, screened articles will need to be identified for inclusion by both reviewers. Conflicts will be resolved by a third independent reviewer not involved in the initial screen. Full-text articles from the screening process will then be read for final inclusion in the review. A PRISMA flow diagram will be used to illustrate the number of studies screened, those chosen for inclusion, as well as reason(s) for exclusion from data extraction after full text review. This diagram will be generated using Covidence automatically after annotating specific studies for inclusion/exclusion.

Data Extraction

Data extraction will be performed independently by two reviewers using Covidence's data extraction tool. We have created a data extraction tool and will test the tool on two papers prior to finalizing the tool's parameters. Conflicts or missing data during data extraction are automatically flagged and will be resolved by a third reviewer when necessary. Data will be extracted with the following data points of interest:

i. **Study Characteristics** – Publication type, Country, Date of publication, study size, and setting (i.e., Medical, Surgical, Cardiovascular, Trauma Centre, etc.), industry sponsorship.

ii. **Population Characteristics** – Age, Sex, Admission Diagnosis, Measurement of Sedation, Overall mortality, Previous diagnosis of cognitive impairment, PTSD, Anxiety, or Depression, previous antipsychotic use.

iii. **Intervention Characteristics -** Inhaled or intravenous agent used; indication for sedation (postoperative cardiac, postoperative non-cardiac, non-surgical, or mixed), concentration (minimum alveolar concentration (MAC) or end-tidal

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concentration for inhaled if available), primary vs adjunct mode of sedation, total duration of exposure to inhaled anesthetic, mode of administration (i.e., Anesthesia Conserving Device (AnaConDa) vs other)

iv. **Comparator Characteristics –** Comparator(s) of interest include intravenous sedative used (e.g., Propofol, opioid, benzodiazepine, ketamine), dose of comparator(s) of interest used and age and sex of comparator group.

v. **Outcomes –** For each of cognitive function, anxiety, depression, and PTSD: tool(s) of measurement (subjective vs objective, and the specific tool (s) used (i.e., questionnaires, Mini Mental State Examination, Trails A, etc. for cognition; Hospital Anxiety and Depression or Patient Health Questionaire-9 for depression, etc. or a battery of tests for each outcome)), definition of impairment (scores on objective testing), timing of measurement (e.g., while in ICU, prior to hospital discharge; 3 months post-discharge, etc.), rates of outcome of interest (e.g., % with depressive symptoms)

   i. For cognitive function only: domain of cognition affected (e.g., executive function, language, etc.).

   ii. For Depression, Anxiety, PTSD only: rates of use of psychiatric medications (e.g., antidepressants/anxiolytics, etc.) in follow-up.

Methodological Appraisal

Once we have selected studies for data extraction, each individual study will also be assessed for risk of bias based on publication type and validated screening tools. In order to assess selected studies for bias, we will use the Cochrane Risk of Bias tool for any randomized controlled trials,

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and report individual domains of bias as either "High" or "Low" risk of bias. To assess biases in case-control and cohort studies, we will use the Newcastle-Ottawa Quality Assessment Scale. For any case series that are included, we will use the Joanna Briggs Institute (JBI) Critical Appraisal tool for case series.

Data synthesis

We will perform a narrative description of included full-text studies for each of the four outcomes of cognition, depression, anxiety, and PTSD. When available, we will present each individual outcome of interest in table format with a descriptive statistical focus with categorical variables (e.g., sex, prevalence of quantitative impairment) expressed as a frequency of events, and quantitative variables from each study presented as medians. We do not anticipate a sufficient number of randomized controlled trials in this area to allow for a meta-analysis to be performed.

Sensitivity analyses

If possible, a sensitivity analysis will be performed on the indication for sedation (cardiac post-operative patients vs non-cardiac postoperative patients vs non-operative), as well as for duration of sedation (for more than 24 hours or less than 24 hours) in order to ascertain the effect of these variables on the outcome of interest.

Patient and Public Involvement

As this is a systematic review protocol, patients and the public were not directly involved in the formation of the protocol. Patient outcomes will be derived from original articles when published in the systematic review.

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Ethics and dissemination: The systematic review uses published data and therefore does not require ethics approval. Results will be disseminated via publication in peer-reviewed journals and presentation at conferences related to the field.

FUNDING

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors

COMPETING INTERESTS

AJ and MS have an ongoing clinical trial (clinicaltrials.gov, NCT04415060) investigating the use of inhaled anesthetics in the ICU. There are no other conflicts of interest (financial or non-financial) to report for the remaining authors.

AUTHOR CONTRIBUTIONS

MS and SC designed the project with assistance from AJ. SC drafted the initial manuscript. KG, CF, DW, LB, and AJ all assisted in project design and writing of the manuscript. All authors edited and approved the final submission.

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Table 1. Inclusion and Exclusion Criteria

POPULATION	Adult (>18 year-old) patients admitted to critical care units
INTERVENTION	Sedation with Inhaled Anesthetics
COMPARATOR	Intravenous sedation
OUTCOME	Short- and Long-term Cognition and Psychiatric outcomes

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Search History saved as "2020.11. SR_volatile anesthetics vs intravenous sedation_cognition_M_FNL"
Search History saved as "2020.11. SR_volatile anesthetics vs intravenous sedation_cognition_E_FNL"
Search History saved as "2020.11. SR_volatile anesthetics vs intravenous sedation_cognition_PsycInfo_FNL"

Topic: Effect of Sedation with Inhaled Anesthetics on Cognitive and Psychiatric Outcomes in Critically Ill Adults: A Systematic Review

For: Dr. Sean Cuninghame; Dr. Kevin Gorsky, Dr. Conall Franceour, Dr. Davinia Withington, Dr. Lisa Burry, Dr. Angela Jerath, Dr. Marat Slessarev,

Database: Ovid MEDLINE(R) ALL <1946 to December 02, 2020>

Search Strategy:

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1 exp Anesthetics, Inhalation/ or Anesthesia, Inhalation/ (60387)

2 (((inhal* or insufflation$ or gas or gaseous$ or volatile$ or halogen$) adj3 (an?esthe* or sedat*)) or (inhalation$ adj3 induction$) or (ftorotan$ or fluothane$ or narcotan$ or isoflurane$ or aerane$ or aerrane$ or forane$ or forene$ or forthane$ or isoflurano$ or isorane$ or sofloran$ or sevocalm$ or sevoflo$ or sevofrane$ or sevohale$ or sevoflurane$ or sevorane$ or sevotec$ or ultane$ or anestane$ or bromochlorotrifluorethane$ or (bromo$ adj2 chloro$ adj2 trifluorethane$) or fluorothan* or halan$ or halothan* or halthane$ or ineltano$ or phthorothane$ or trothane$ or desflurane$ or enflurane$ or penthrox$)).tw,kf. (42998)

3 or/1-2 [volatile anesthetics concept] (76374)

4 Critical Care/ or Critical Illness/ or exp Intensive Care Units/ (145037)

5 ((intens* or critical*) adj3 (care or ill or illness or unit*)).tw,kf. (212397)

6 (ICU or ICUs or CCU or CCUs or ITU or HDU).tw,kf. (65415)

7 or/4-6 [ICU concept] (276089)

8 (intens$ adj3 care).af. or (intensive* or critical$).jw,ja,jn. or (sedat$.tw,kf. and (exp *Respiration, Artificial/ or (((ventilat$ or respirat$) adj2 (artificial$ or mechanical$ or pneumon$ or controlled)) or (ventilat$ adj3 patient$)).tw,kf.)) (360980)

9 7 or 8 (429901)

10 3 and 9 (4309)

11 Anesthesia, Intravenous/ or Infusions, Intravenous/ or (intravenous$ or intra-venous$).mp. or (IV or IVs or TIVA or Auto?TIVA or Manual?TIVA).tw,kf. (804110)

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12 Propofol/ or (propofol$ or diprivan$).tw,kf. or Ketamine/ or (ketamin$ or esketamin$ or ketanest$ or ketalar$ or ketaset$).tw,kf. or Midazolam/ or (midazolam$ or dormicum$ or Versed$ or buccolam$ or dalam or dormonid or fortanest or fulsed or hypnovalor hypnovel or hypnoyvel or ipnovel or midacum or midazo or midazol or midolamor miloz).tw,kf. or conscious sedation/ or deep sedation/ or ((conscious$ or moderate$ or deep$) adj2 sedat$).tw,kf. (62360)

13 12 and (continuous$ or prolonged$ or concentration$ or infusion$).tw,kf. (18739)

14 11 or 13 [Intravenous Anesthesia OR Midazolam/Ketamine/Propofol/ continuous administration - concept] (813003)

15 10 and 14 (1475)

16 exp Cognition Disorders/ or exp Cognition/ or exp Cognitive Therapy/ or Cognitive Aging/ or Tremor/ or exp neuropsychological tests/ (409458)

17 neurobehavioral manifestations/ or exp communication disorders/ or exp memory disorders/ or intellectual disability/ or exp perceptual disorders/ or exp sensation disorders/ (355136)

18 (cognition$ or cognitive$ or neuropsycholog$ or neuro-psycholog$ or neurocognit$ or neuro?affectiv$ or neuro-cognit$ or neurobehav$ or neuro-behav$ or aphasia$ or (involuntary$ adj2 quiver$) or tremor$).mp. (575907)

19 exp *Multiple Sclerosis/ or *Alzheimer Disease/ or exp *parkinsonian disorders/ or delirium/ or (delirium$ or delirious$ or Parkinson$ or Alzheimer$).tw. or ((multiple$ or disseminated$) adj5 sclerosis$).tw,kw. (358897)

20 or/16-19 (1269102)

21 exp Prefrontal Cortex/pa, pp or exp Cerebral Cortex/pa, pp or exp Hippocampus/pa, pp or Cerebral Infarction/pa, pp or White Matter/pa, pp or Brain Diseases/pa, pp or Subthalamic Nucleus/pa, pp or exp Subarachnoid Hemorrhage/pa, pp (128097)

22 (hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-thalam$ or intracranial$ or intra-cranial$).tw,kw. (1015992)

23 ((cortex$ or cortices$ or area$ or region$) adj3 (orbital$ or prefrontal$ or orbitofrontal$)).tw,kw. (54920)

24 ((pars$ adj (triangulari$ or operculari$)) or ((area$ or region$) adj2 (speech$ or language$ or broca$)) or broca's or brocas or subcallosal$).tw,kw. (6116)

25 ((frontal$ or olfactory$) adj3 (sulci or (superior$ adj5 convolution$) or sulcus$)).tw,kw. (665)

26 ((frontal$ or orbital$ or orbitofrontal$ or rectal or rectus or straight or marginal) adj3 (gyri$ or gyrus$)).tw,kw. (11762)

27 or/21-26 (1115455)

28 (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$).tw,kw. (1691979)

29 27 and 28 (291070)

30 ((brain or cortex$ or cortices$ or cortical$ or subcortical$ or sub-cortical$ or hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-

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thalam$ or intracranial$ or intra-cranial$) adj2 (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$ or impair$ or decline$ or defect$ or lesion$ or disorder$ or patholog$ or integrity or hyperintens$)).tw,kw. (256693)

31 ((visuospatial$ or (visuo$ adj2 spatial$) or (verbal$ adj2 (comm$ or fluen$)) or (process$ adj2 speed$) or psychomotor$ or psycho-motor$ or memory$ or personality$ or perceptual$ or attention$ or concentration$ or auditory$ or visual$ or sensory$ or sensation$ or spatial$ or IQ or intelligence$ or intellectual$) adj5 (impairment$ or impaired or deficit$ or deficien$ or abnormal$ or dysfunction$ or function$ or skill$ or problem$ of complain$)).mp. (285691)

32 (MMSE or (mini?mental$ adj3 stat$ adj3 exam$)).tw,kf. (11443)

33 exp Stress, Psychological/ or anxiety/ or Depression/ or depressive disorder/ or depressive disorder, major/ or depressive disorder, treatment-resistant/ or (stress$ or ((hospital or scale$ or score$ or questionnaire$) and (HADS-D or PHQ-9 or PHQ or HADS)) or depress$ or anxiety$).tw,kf. (1471121)

34 Stress Disorders, Post-Traumatic/ or (shock$ adj2 shell$).tw,kf. or ((posttrauma$ or (post$ adj3 traumatic$) or post-trauma$) adj3 (stress$ or disorder$ or psychiatric$)).tw,kf. or ((acute$ adj2 stress$ adj2 disorder$) and ASD).tw,kf. (47060)

35 ((posttraumatic$ or (post$ adj3 traumatic$) or post-traumatic$ or stress$ or disorder$) and PTSD).tw,kf. (24021)

36 or/33-35 (1478446)

37 blood pressure/ or heart rate/ or ((heart$ or pulse$) adj3 rate$1).tw. or (cardiac$ adj2 chronotrop$).tw. or (pressure$ adj3 (blood$ or diastolic$ or pulse$ or systolic$)).tw. (611661)

38 (psychiatr$ or disability$).tw,kf. or exp Intellectual Disability/ or exp Disability Evaluation/ or "International Classification of Functioning, Disability and Health"/ (522184)

39 or/37-38 (1128539)

40 20 or 29 or 30 or 31 or 32 or 36 or 39 [Cognitive concept] (3749930)

41 10 and 40 [ICU concept PLUS volatile anesthetics concept PLUS Cognitive concept] (1299)

42 exp Survivors/ or surviv$.mp. or (post?intensive$ or postdischarg$ or ((post$ or after$) adj3 (critical$ or ICU or intensive$ or discharg$))).tw,kf. or Comparative Study/ or (vs$1 or vs or versus or compar$ or during-and-after).tw. or ((multicenter$ or multi-center$).tw. and ((between or each or both or either or than or across) adj2 group$1).tw,kw.) or groups.tw. or group$1.ab. /freq=2 (9075813)

43 ((random$ or multicenter$ or multi-center$ or center$ or multicentr$ or multi-centr$ or centr$ or pilot$ or population$ or observational or validation or phase$) adj3 (trial or study)).mp. or clinical trial.tw. or exp clinical trial/ or double-blind method/ or random allocation/ (1809348)

44 or/42-43 [Random/Clinical trial / Survivors / Comparative Study] (9679756)

45 10 and 14 and 44 [WITHOUT Cognitive concept PRESENT could be found in the Full Text - ICU concept PLUS volatile anesthetics concept PLUS Intravenous Anesthesia concept PLUS Random/Clinical trial / Survivors / Comparative Study] (1095)

46 41 or 45 (1915)

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47 46 not (exp Animals/ not (Human/ and exp Animals/)) (1511)

48 47 not (animal$1 or mice or rat or rats or cat$1 or cattle$1 or dog$1 or goat$1 or horse$1 or rabbit$1 or sheep$1 or swine$1 or pig$1 or piglet* or canine$1 or feline$1 or porcine$ or calf or primate* or rodent$ or hamster$ or lamb$1 or monkey$1 or murine or veterinar*).ti. (1480)

49 limit 48 to "all adult (19 plus years)" (863)

50 limit 48 to "all child (0 to 18 years)" (425)

51 48 not (50 not (49 and 50)) (1205)

52 51 and adult$.ti. (32)

53 51 and (pediatr$ or paediatr$ or child$ or adolescent$ or infan$ or newborn$ or boy$1 or neonat$).ti. (23)

54 51 not (53 not (52 and 53)) (1184)

55 exp case-control studies/ or (case$ and control$).tw. or (case$ and series).tw. [Medline case series] (1623777)

56 case reports/ or case report$.mp. (2227506)

57 54 not (56 not (55 and 56)) (1117)

58 57 not case report.ti. [Removing case Reports and retaining Case Series] (1113)

59 limit 58 to english language (1038)

60 limit 59 to yr="1970 -Current" (1037)

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Database: Embase Classic+Embase <1947 to 2020 December 02>

Search Strategy:

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1 exp *inhalation anesthetic agent/ or exp inhalation anesthesia/ (73272)

2 (((inhal* or insufflation$ or gas or gaseous$ or volatile$ or halogen$) adj3 (an?esthe* or sedat*)) or (inhalation$ adj3 induction$) or (ftorotan$ or fluothane$ or narcotan$ or isoflurane$ or aerane$ or aerrane$ or forane$ or forene$ or forthane$ or isoflurano$ or isorane$ or sofloran$ or sevocalm$ or sevoflo$ or sevofrane$ or sevohale$ or sevoflurane$ or sevorane$ or sevotec$ or ultane$ or anestane$ or bromochlorotrifluorethane$ or (bromo$ adj2 chloro$ adj2 trifluorethane$) or fluorothan* or halan$ or halothan* or halthane$ or ineltano$ or phthorothane$ or trothane$ or desflurane$ or enflurane$ or penthrox$)).tw,kw. (59537)

3 or/1-2 [volatile anesthetics concept] (100064)

4 critically ill patient/ or exp intensive care unit/ or intensive care/ or critical illness/ (351059)

5 ((intens* or critical*) adj3 (care or ill or illness or unit*)).tw,kw. (322355)

6 (ICU or ICUs or CCU or CCUs or ITU or HDU).tw,kw. (133390)

7 or/4-6 [ICU concept] (483378)

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8 (intensive* or critical$).ja,jn,jx. or (sedat$.tw,kw. and (exp *artificial ventilation/ or (((ventilat$ or respirat$) adj2 (artificial$ or mechanical$ or pneumon$ or controlled)) or (ventilat$ adj3 patient$)).tw,kw.)) (267016)

9 7 or 8 (640893)

10 3 and 9 (4189)

11 exp intravenous anesthesia/ or exp intravenous drug administration/ or (intravenous$ or intra-venous$).mp. or (IV or IVs or TIVA or Auto?TIVA or Manual?TIVA).tw,kw. (1724896)

12 Propofol/ or (propofol$ or diprivan$).tw,kw. or Ketamine/ or (ketamin$ or esketamin$ or ketanest$ or ketalar$ or ketaset$).tw,kw. or Midazolam/ or (midazolam$ or dormicum$ or Versed$ or buccolam$ or dalam or dormonid or fortanest or fulsed or hypnovalor hypnovel or hypnoyvel or ipnovel or midacum or midazo or midazol or midolamor miloz).tw,kw. or conscious sedation/ or deep sedation/ or ((conscious$ or moderate$ or deep$) adj2 sedat$).tw,kw. (137818)

13 12 and (continuous$ or prolonged$ or concentration$ or infusion$).tw,kw. (34730)

14 11 or 13 [Intravenous Anesthesia OR Midazolam/Ketamine/Propofol/ continuous administration - concept] (1741681)

15 10 and 14 (1848)

16 exp *cognitive defect/ or exp *Cognition/ or exp *Cognitive Therapy/ or exp *neuropsychological test/ or experimental cognitive test/ (1135100)

17 exp *communication disorder/ or exp *memory disorders/ or exp *perception disorder/ or exp *sensory dysfunction/ (288272)

18 (cognition$ or cognitive$ or neuropsycholog$ or neuro-psycholog$ or neurocognit$ or neuro?affectiv$ or neuro-cognit$ or neurobehav$ or neuro-behav$ or aphasia$ or (involuntary$ adj2 quiver$) or tremor$).mp. (866016)

19 *Multiple Sclerosis/ or *Alzheimer Disease/ or parkinsonism/ or exp delirium/ or (delirium$ or delirious$ or Parkinson$ or Alzheimer$).tw. or ((multiple$ or disseminated$) adj5 sclerosis$).tw,kw. (544154)

20 or/16-19 (2199890)

21 (hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-thalam$ or intracranial$ or intra-cranial$).tw,kw. (1452911)

22 ((cortex$ or cortices$ or area$ or region$) adj3 (orbital$ or prefrontal$ or orbitofrontal$)).tw,kw. (77598)

23 ((pars$ adj (triangulari$ or operculari$)) or ((area$ or region$) adj2 (speech$ or language$ or broca$)) or broca's or brocas or subcallosal$).tw,kw. (8645)

24 ((frontal$ or olfactory$) adj3 (sulci or (superior$ adj5 convolution$) or sulcus$)).tw,kw. (877)

25 ((frontal$ or orbital$ or orbitofrontal$ or rectal or rectus or straight or marginal) adj3 (gyri$ or gyrus$)).tw,kw. (16960)

26 or/21-25 (1519587)

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27 brain atrophy/ or exp *brain disease/ or (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$).tw,kw. (3600056)

28 26 and 27 (756089)

29 ((brain or cortex$ or cortices$ or cortical$ or subcortical$ or sub-cortical$ or hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-thalam$ or intracranial$ or intra-cranial$) adj2 (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$ or impair$ or decline$ or defect$ or lesion$ or disorder$ or patholog$ or integrity or hyperintens$)).tw,kw. (387036)

30 ((visuospatial$ or (visuo$ adj2 spatial$) or (verbal$ adj2 (comm$ or fluen$)) or (process$ adj2 speed$) or psychomotor$ or psycho-motor$ or memory$ or personality$ or perceptual$ or attention$ or concentration$ or auditory$ or visual$ or sensory$ or sensation$ or spatial$ or IQ or intelligence$ or intellectual$) adj5 (impairment$ or impaired or deficit$ or deficien$ or abnormal$ or dysfunction$ or function$ or skill$ or problem$ of complain$)).tw,kw. (371991)

31 (MMSE or (mini?mental$ adj3 stat$ adj3 exam$)).tw,kw. (24495)

32 mental stress/ or anxiety/ or Depression/ or major depression/ or treatment resistant depression/ or (stress$ or ((hospital or scale$ or score$ or questionnaire$) and (HADS-D or PHQ-9 or PHQ or HADS)) or depress$ or anxiety$).tw,kw. (2068051)

33 posttraumatic stress disorder/ or (shock$ adj2 shell$).tw,kw. or ((posttrauma$ or (post$ adj3 traumatic$) or post-trauma$) adj3 (stress$ or disorder$ or psychiatric$)).tw,kw. or ((acute$ adj2 stress$ adj2 disorder$) and ASD).tw,kw. (69838)

34 ((posttraumatic$ or (post$ adj3 traumatic$) or post-traumatic$ or stress$ or disorder$) and PTSD).tw,kw. (31596)

35 or/32-34 (2079206)

36 exp *blood pressure/ or heart rate/ or heart rate variability/ or respiratory sinus arrhythmia/ or resting heart rate/ or ((heart$ or pulse$) adj3 rate$1).tw. or (cardiac$ adj2 chronotrop$).tw. or (pressure$ adj3 (blood$ or diastolic$ or pulse$ or systolic$)).tw. (799450)

37 (psychiatr$ or disability$).tw,kw. or intellectual impairment/ or exp mental deficiency/ or exp *disability/ or "International Classification of Functioning, Disability and Health"/ (777645)

38 or/36-37 (1568243)

39 20 or 28 or 29 or 30 or 31 or 35 or 38 [Cognitive concept] (5794698)

40 10 and 39 [ICU concept PLUS volatile anesthetics concept PLUS Cognitive concept] (1385)

41 survivor/ or surviv$.mp. or (post?intensive$ or postdischarg$ or ((post$ or after$) adj3 (critical$ or ICU or intensive$ or discharg$))).tw,kw. or exp comparative study/ or (vs$1 or vs or versus or compar$ or during-and-after).tw. or ((multicenter$ or multi-center$).tw. and ((between or each or both or either or than or across) adj2 group$1).tw,kw.) or groups.tw. or group$1.ab. /freq=2 (12126728)

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42 ((random$ or multicenter$ or multi-center$ or center$ or multicentr$ or multi-centr$ or centr$ or pilot$ or population$ or observational or validation or phase$) adj3 (trial or study)).mp. or clinical trial.tw. or exp clinical study/ or randomization/ (11071495)

43 or/41-42 [Random/Clinical trial / Survivors / Comparative Study] (18293986)

44 10 and 14 and 43 [WITHOUT Cognitive concept PRESENT could be found in the Full Text - ICU concept PLUS volatile anesthetics concept PLUS Intravenous Anesthesia concept PLUS Random/Clinical trial / Survivors / Comparative Study] (1514)

45 40 or 44 (2262)

46 45 not ((exp animal/ or nonhuman/) not exp human/) (1900)

47 46 not (animal$1 or mice or rat or rats or cat$1 or cattle$1 or dog$1 or goat$1 or horse$1 or rabbit$1 or sheep$1 or swine$1 or pig$1 or piglet* or canine$1 or feline$1 or porcine$ or calf or primate* or rodent$ or hamster$ or lamb$1 or monkey$1 or murine or veterinar*).ti. (1855)

48 limit 47 to (adult <18 to 64 years> or aged <65+ years>) (767)

49 limit 47 to (embryo or infant or child) (326)

50 47 not (49 not (48 and 49)) (1578)

51 50 and adult$.ti. (24)

52 50 and (pediatr$ or paediatr$ or child$ or adolescent$ or infan$ or newborn$ or boy$1 or neonat$).ti. (81)

53 50 not (52 not (51 and 52)) (1500)

54 exp case control study/ or (case$ and control$).tw. or exp case study/ or (case$ and series).tw. [EMBASE case series] (1136668)

55 case report/ or case report$.tw,kw. (2743289)

56 53 not (55 not (54 and 55)) (1216)

57 56 not case report.ti. [Removing case Reports and retaining Case Series] (1202)

58 limit 57 to english language (1078)

59 limit 58 to yr="1970 -Current" (1068)

60 limit 59 to embase (668)

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Database: APA PsycInfo <1967 to November Week 4 2020>

Search Strategy:

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1 (((inhal* or insufflation$ or gas or gaseous$ or volatile$ or halogen$) adj3 (an?esthe* or sedat*)) or (inhalation$ adj3 induction$) or (ftorotan$ or fluothane$ or narcotan$ or isoflurane$ or aerane$ or aerrane$ or forane$ or forene$ or forthane$ or isoflurano$ or isorane$ or sofloran$ or sevocalm$ or sevoflo$ or sevofrane$ or sevohale$ or sevoflurane$ or sevorane$ or sevotec$ or ultane$ or anestane$

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or bromochlorotrifluorethane$ or (bromo$ adj2 chloro$ adj2 trifluorethane$) or fluorothan* or halan$ or halothan* or halthane$ or ineltano$ or phthorothane$ or trothane$ or desflurane$ or enflurane$ or penthrox$)).tw. [volatile anesthetics concept] (1319)

2 intensive care/ (4348)

3 ((intens* or critical*) adj3 (care or ill or illness or unit*)).tw. (13241)

4 (ICU or ICUs or CCU or CCUs or ITU or HDU).mp. (2736)

5 or/2-4 [ICU concept] (14101)

6 (intens$ adj3 care).af. or (intensive* or critical$).jn,jx. or (sedat$.mp. and (*artificial respiration/ or (((ventilat$ or respirat$) adj2 (artificial$ or mechanical$ or pneumon$ or controlled)) or (ventilat$ adj3 patient$)).tw.)) (38989)

7 5 or 6 (42014)

8 1 and 7 (94)

9 intravenous injections/ or (intravenous$ or intra-venous$).mp. or (IV or IVs or TIVA or Auto?TIVA or Manual?TIVA).tw. (76553)

10 Propofol/ or (propofol$ or diprivan$).mp. or Ketamine/ or (ketamin$ or esketamin$ or ketanest$ or ketalar$ or ketaset$).mp. or Midazolam/ or (midazolam$ or dormicum$ or Versed$ or buccolam$ or dalam or dormonid or fortanest or fulsed or hypnovalor hypnovel or hypnoyvel or ipnovel or midacum or midazo or midazol or midolamor miloz).mp. or conscious sedation/ or deep sedation/ or ((conscious$ or moderate$ or deep$) adj2 sedat$).tw. (6565)

11 10 and (continuous$ or prolonged$ or concentration$ or infusion$).mp. (1488)

12 9 or 11 [Intravenous Anesthesia OR Midazolam/Ketamine/Propofol/ continuous administration - concept] (77551)

13 8 and 12 (30)

14 exp Cognition/ or exp Cognitive Therapy/ or Tremor/ or exp neuropsychological assessment/ (69761)

15 exp communication disorders/ or exp memory disorders/ or cognitive impairment/ or exp perceptual disturbances/ (112915)

16 (cognition$ or cognitive$ or neuropsycholog$ or neuro-psycholog$ or neurocognit$ or neuro?affectiv$ or neuro-cognit$ or neurobehav$ or neuro-behav$ or aphasia$ or (involuntary$ adj2 quiver$) or tremor$).mp. (663872)

17 *Multiple Sclerosis/ or *alzheimer's disease/ or parkinsonism/ or delirium/ or (delirium$ or delirious$ or Parkinson$ or Alzheimer$).tw. or ((multiple$ or disseminated$) adj5 sclerosis$).tw. (114306)

18 or/14-17 (780410)

19 (hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-thalam$ or intracranial$ or intra-cranial$).tw. (201667)

20 ((cortex$ or cortices$ or area$ or region$) adj3 (orbital$ or prefrontal$ or orbitofrontal$)).tw. (40292)

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21 ((pars$ adj (triangulari$ or operculari$)) or ((area$ or region$) adj2 (speech$ or language$ or broca$)) or broca's or brocas or subcallosal$).tw. (5216)

22 ((frontal$ or olfactory$) adj3 (sulci or (superior$ adj5 convolution$) or sulcus$)).tw. (366)

23 ((frontal$ or orbital$ or orbitofrontal$ or rectal or rectus or straight or marginal) adj3 (gyri$ or gyrus$)).tw. (7911)

24 or/19-23 (238556)

25 (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$).tw. (282352)

26 24 and 25 (66855)

27 ((brain or cortex$ or cortices$ or cortical$ or subcortical$ or sub-cortical$ or hippocamp$ or (white$ adj2 matter$) or neurologic$ or cerebr$ or subarachnoid$ or sub-arachnoid$ or subthalam$ or sub-thalam$ or intracranial$ or intra-cranial$) adj2 (deficit$ or dysfunc$ or atrophy or h?emorrhag$ or abnormal$ or abscess$ or impair$ or decline$ or defect$ or lesion$ or disorder$ or patholog$ or integrity or hyperintens$)).tw. (64349)

28 ((visuospatial$ or (visuo$ adj2 spatial$) or (verbal$ adj2 (comm$ or fluen$)) or (process$ adj2 speed$) or psychomotor$ or psycho-motor$ or memory$ or personality$ or perceptual$ or attention$ or concentration$ or auditory$ or visual$ or sensory$ or sensation$ or spatial$ or IQ or intelligence$ or intellectual$) adj5 (impairment$ or impaired or deficit$ or deficien$ or abnormal$ or dysfunction$ or function$ or skill$ or problem$ of complain$)).mp. (179366)

29 (MMSE or (mini?mental$ adj3 stat$ adj3 exam$)).tw. (6045)

30 exp stress/ or exp anxiety/ or exp major depression/ or (stress$ or ((hospital or scale$ or score$ or questionnaire$) and (HADS-D or PHQ-9 or PHQ or HADS)) or depress$ or anxiety$).tw. (637796)

31 exp posttraumatic stress disorder/ or (shock$ adj2 shell$).tw. or ((posttrauma$ or (post$ adj3 traumatic$) or post-trauma$) adj3 (stress$ or disorder$ or psychiatric$)).tw. or ((acute$ adj2 stress$ adj2 disorder$) and ASD).tw. (46884)

32 ((posttraumatic$ or (post$ adj3 traumatic$) or post-traumatic$ or stress$ or disorder$) and PTSD).tw. (32616)

33 or/30-32 (640126)

34 blood pressure/ or heart rate/ or ((heart$ or pulse$) adj3 rate$1).tw. or (cardiac$ adj2 chronotrop$).tw. or (pressure$ adj3 (blood$ or diastolic$ or pulse$ or systolic$)).tw. (39341)

35 (psychiatr$ or disability$).mp. or cognitive impairment/ or disability evaluation/ (428052)

36 or/34-35 (464663)

37 18 or 26 or 27 or 28 or 29 or 33 or 36 [Cognitive concept] (1633003)

38 8 and 37 [ICU concept PLUS volatile anesthetics concept PLUS Cognitive concept] (53)

39 38 or 13 (62)

40 39 not (animal$1 or mice or rat or rats or cat$1 or cattle$1 or dog$1 or goat$1 or horse$1 or rabbit$1 or sheep$1 or swine$1 or pig$1 or piglet* or canine$1 or feline$1 or porcine$ or calf or primate* or rodent$ or hamster$ or lamb$1 or monkey$1 or murine or veterinar*).ti. (44)

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41 40 and adult$.ti. (1)

42 40 and (pediatr$ or paediatr$ or child$ or adolescent$ or infan$ or newborn$ or boy$1 or neonat$).ti. (5)

43 40 not (42 not (41 and 42)) (39)

44 limit 43 to english language (36)

45 limit 44 to yr="1970 -Current" (36)

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